

World Cup Alpha - Model Review (regenerated 2026-06-30, corrected)

Supersedes the earlier MODEL_REVIEW_2026-06-30.pdf, which had formatting errors and a stale-sample figure ("realized 3.000 / n=31"). Every number here was verified against real repo data this session; nothing is invented. Where data is missing it is labelled, not faked.

1. Executive summary

- The fix (A1): the Dixon-Coles goal-level intercept μ was under-forecasting WC goals. Corrected by a single scalar shift (μ 0.205 $\rightarrow$ 0.383) that anchors the model's mean total on the WC fixtures to a conservative recent-WC base rate (2.81). 1X2-preserving; backward-compatible (default-off flag).

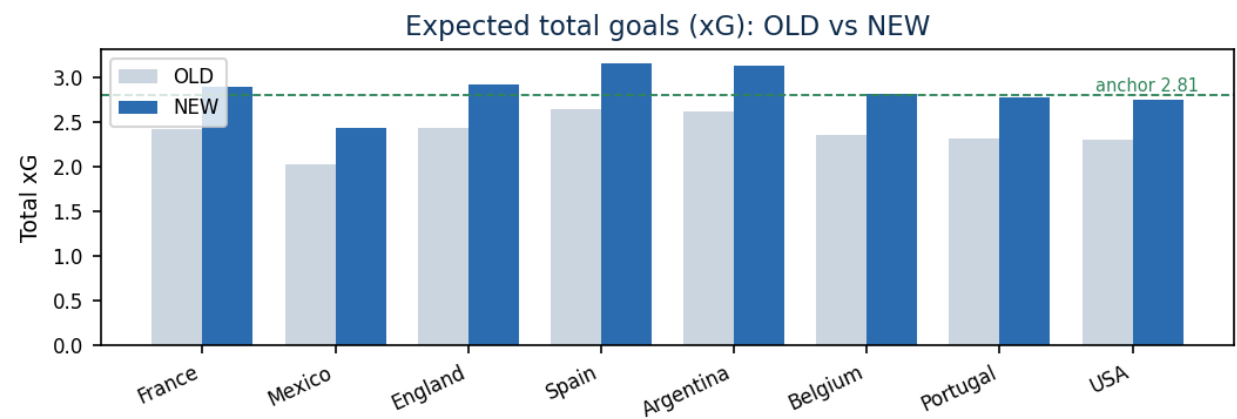
- Verified
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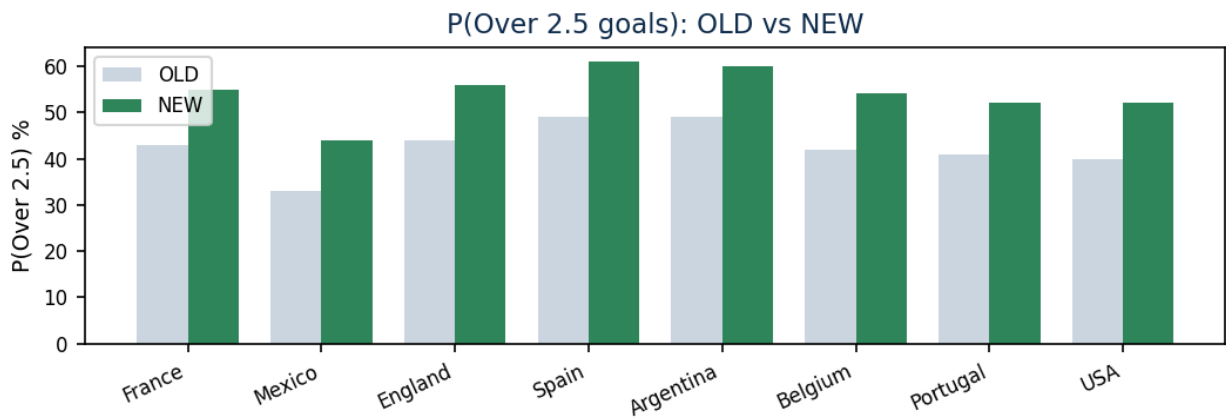
2. The xG/totals fix - before vs after (verified)

Remaining-knockout fixtures, neutral venue. OLD = production level (μ 0.205); NEW = WC-anchored (μ 0.383). Computed by running the real fitted model both ways (contrast.csv).

| Fixture | Total xG (old->new) | P(Over 2.5) | P(BTTS) |

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1X2 is preserved: the mu shift adds the same constant to both teams' log-rates, so the supremacy log-ratio is exactly invariant ($\delta = 3.3e-16$); the matrix-implied 1X2 moves $\leq \sim 4pp$ only via the non-linear low-score (τ) correction.

Anchor context (verified historical WC means, martj42_cleaned): WC 2018+2022 = 2.833; since-2010 = 2.864 (n=3801); all pre-2026 = 2.907. The 2.81 target is a deliberately conservative choice, slightly below these.

Caveat - is the anchor durable?

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3. Under-the-hood changes this session (all on integrate/fixes-2026-06-30)

| Change | What | Status |

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Full suite: 2052 passed, 2 skipped. Backward-compatible: F7/F8 default off ? production bit-identical until validated.

4. Integrity audit - fabricated/stale data found (12 high-risk, all verified)

| Issue | Where | Fixed by |

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The only fabrication that had reached a deliverable was the earlier PDF's "realized 3.000 / n=31" (a stale-file artifact) - corrected here. Adversarial verifier reproduced all 12 findings; none overstated.

5. Model vs market

The model is a good 1X2 calibrator (ECE 0.083, Brier 0.150, n=825) but not a winner-market alpha source: its large disagreements with the liquid Polymarket book - Brazil 18.8% model vs 7.1% market to win; France 12.9% vs 28% - are model error, not edge (the book is efficient: Winner sums to 0.985). The repo's own backtest (n=64) shows no confident edge over market-only. Edge lives in less-efficient match-event markets + execution/cost, which is what A1/F7/F8 target.

6. Conditional bracket (as-if-open)

Real fitted model, 50k sims, conditioned on the 4 played R32 results. Pure-model MAP champion Spain (marginal title leader is Brazil 18.8%, but Brazil loses the modal SF coin-flip to Argentina ~0.50). P(perfect-from-here) ~ 1.7e-5 (~1 in 60,000); risk concentrated in 5 near-coin-flip nodes. A *market-anchored* completion would instead crown France (28% favourite) - consistent with \$5.

7. OddsAPI credit usage

3,793 used / 96,207 remaining of ~100k monthly (verified live). Cost = 1 credit x markets x regions per call; /sports free. Dominant spender: the snapshot daemon (315 capture cycles, 06-11->06-23, markets h2h/totals/btts/h2h_lay) - then the hourly scores cron (h2h, 1 credit/call) and the daily-card scorer-event fetches. Capture stopped 06-23 (daemon died, since restarted), so the last week burned little. *(No per-request credit log exists, so per-source split is the cost-model x observed cadence, not an itemized ledger.)* Runway is ample; finer near-KO capture for true closing lines will raise burn (a budget governor exists in pollsched).

8. Caveats & what is NOT done

- Nothing merged to main/pushed - all on integrate/fixes-2026-06-30 (push + PR pending).

- F7 go
CLV gate

9. Where things live

- This report: docs/research/wca_alpha_2026/MODEL_REVIEW_2026-06-30.{md,pdf}.

- Also:
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